Debugging shapes and gradients

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Give each check a specific error to expose.

Show complete arrays,minus/plus probes and fixed-state updates;distinguish a removed slot from a zeroed bias term.

Keep one habit: Assess independent coordinate repair and checked loss with the supplied key;retain failures and regression tests.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Check dimension **contract**s before computing a layer

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

A deterministic error can repeat perfectly. Independent expected values and checks of intermediate computations are needed.

Opening retrieval

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

Look for: Trace how starting values affect a fixed **training** budget

Reference example, not the only acceptable wording: Teacher explanation: States are -2, 0 and 1. Final **training** MSE is 2.5. Each step halves the signed distance from 2.

Without opening the old lesson, explain one key idea from **Training** and validation curves. Give an example and a mistake that the idea helps you avoid.

Look for: Compute comparable **training** and validation losses at saved checkpoints

Reference example, not the only acceptable wording: Teacher explanation: **Prediction**s [3, 4], residuals [-1.5, -2], squares [2.25, 4], MSE 3.125.

Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

Look for: Trace repeated forward loss gradient and update steps

Reference example, not the only acceptable wording: Teacher explanation: **Prediction**s [0, 0], errors [-2, -4], MSE 10, gradient -10 and new v = 1.

Pictured end goal

By the end, I can explain debugging **shape**s and gradients using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Initialisation and reproducibility

Training and validation curves

Training a small network

For the compact alternative schedule, use two 45-minute sessions. Start with an axis **contract** and a deliberately wrong intermediate value. Keep the dense-layer example separate from the scalar gradient diagnostic.

The central assessment is locating the first error and preserving meaningful **regression check**s, not merely getting a program to execute.

Outcomes

Check dimension **contract**s before computing a layer

Locate the first wrong value in a forward or gradient trace

Verify a repair with independent checks and limited claims

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach shape contracts,intermediates and gradient labels together

> Keep the dense layer,pictured[a,c]model,diagnostic[w,b]model and independent case distinct.

Main convention,complete bias audit and guided key

For this lesson, a dense layer stores one **output** unit per weight-matrix row. If an **input** has two features and the layer has three units, W has three rows and two columns, b has length 3, and one **input** x has length 2.

Compute z for each unit, then apply ReLU to each component.

Use W rows [1, 0], [0, 1], [1, -1] and b = [0, 0, 1]. For x = [1, 2], the scores are [1, 2, 0] and the activations are [1, 2, 0].

A batch stores examples in rows: **input**s [[1, 2], [3, 1]] have **shape** 2 by 2, and **output**s have **shape** 2 by 3.

Other software may use a transposed weight convention. The key is to record and follow one **contract**. An array with the right number of values can still be wrong if features or axes are swapped.

Python zip stops when its shortest **input** ends. Without a length check, a missing feature can silently remove a contribution.

Reject malformed dimensions before computing: do not pad, truncate or re**shape** a mismatched **input** just to make the calculation run.

The function assumes finite numeric values; it checks dimensions, not every possible **input** fault. Evaluate the two batch rows independently and stack their **output**s.

Retain tests that reject a missing **input** feature, a short weight row and an incorrect bias count.

Input; Wrows[1, 0]/[0, 1]/[1, −1] · Correct scores withb[0, 0, 1] · Correct ReLU · Scores ifb 3 set0 · ReLU ifb 3 set0 · What the check reveals

> **Input; Wrows[1, 0]/[0, 1]/[1, −1]:** [1, 2]
>
> **Correct scores withb[0, 0, 1]:** [1, 2, 0]

> **Correct ReLU:** [1, 2, 0]
>
> **Scores ifb 3 set0:** [1, 2, –1]
>
> **ReLU ifb 3 set0:** [1, 2, 0]
>
> **What the check reveals:** Score error hidden by activation

> **Input; Wrows[1, 0]/[0, 1]/[1, –1]:** [0, 2]
>
> **Correct scores withb[0, 0, 1]:** [0, 2, –1]
>
> **Correct ReLU:** [0, 2, 0]
>
> **Scores ifb 3 set0:** [0, 2, –2]
>
> **ReLU ifb 3 set0:** [0, 2, 0]
>
> **What the check reveals:** Again same final activation despite wrong score

> **Input; Wrows[1, 0]/[0, 1]/[1, –1]:** [3, 1]
>
> **Correct scores withb[0, 0, 1]:** [3, 1, 3]
>
> **Correct ReLU:** [3, 1, 3]
>
> **Scores ifb 3 set0:** [3, 1, 2]
>
> **ReLU ifb 3 set0:** [3, 1, 2]
>
> **What the check reveals:** Positive third score exposes the bias bug

In the numeric “omit the third bias”probes, keep its slot and change b=[0, 0, 1]to[0, 0, 0]. That creates a semantic arithmetic bug while all lengths still match. Removing the third slot instead makes bias length 2 and is a different malformed-input case that the guard rejects. The batch contains two row examples and produces a2×3 activation table, not a3×2 table.

Dimension checks do not validate finite numeric values, feature order or intended units.

Guided: correct scores [0, 2, -1], activations [0, 2, 0]. With the third bias omitted, scores become [0, 2, -2], but activations stay [0, 2, 0]. Exit: neither check is sufficient.

Verify the intermediate score, coordinate meanings and expected update, use multiple independent **input**s and retain malformed-dimension checks.

Complete pictured formula and labelled minus/plus probes

Catch a gradient -order bug before the optimiser changes parameters.

Toy y=2a+3c, target 4, start [a, c]=[1, 0]. Loss (y-4)². Program claims gradient [-12,-8] in order [a, c].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate scalar gradient model:y=2a+3c, target 4, old[a, c]=[1, 0], y2, residual−2. Correct ordered gradient[−8, −12]is mislabeled by a program returning[−12, −8]under[a, c]. With spacing.1, a-minus/plus losses 4.84/3.24 give−8; c-minus/plus 5.29/2.89 give−12. Rate.01 with the correct old-state gradient gives[1.08, .12], y2.52, loss 2.1904.

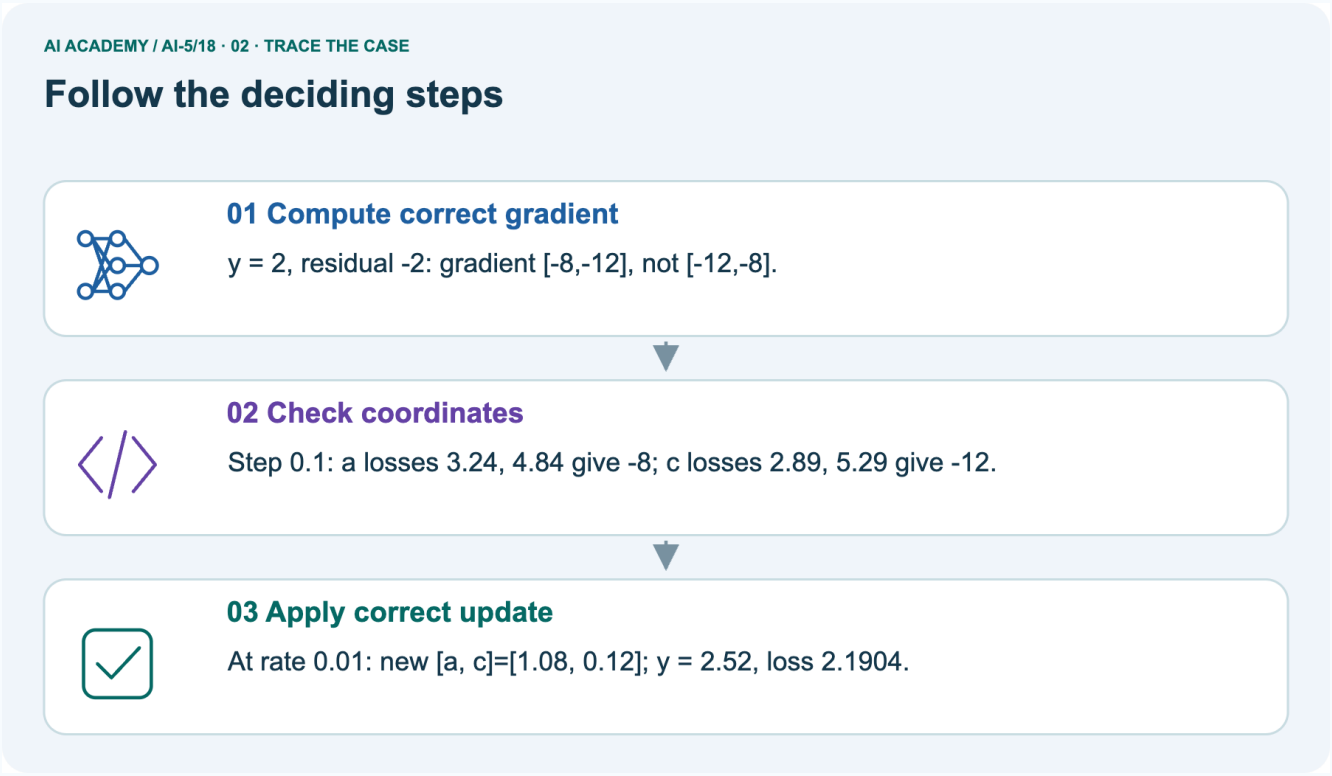
Under a genuinely changed order[c, a], the same gradient is correctly represented[−12, −8], provided every consumer follows that order.

Pictured coordinate probe · Held coordinate · Minus vector /loss · Plus vector /loss · Difference over.2

> **Pictured coordinate probe:** a
>
> **Held coordinate:** c0
>
> **Minus vector /loss:** [.9, 0]/4.84
>
> **Plus vector /loss:** [1.1, 0]/3.24
>
> **Difference over.2:** -8

> **Pictured coordinate probe:** c
>
> **Held coordinate:** a1
>
> **Minus vector /loss:** [1, −.1]/5.29
>
> **Plus vector /loss:** [1, .1]/2.89
>
> **Difference over.2:** -12

Trace semantic repair and correctly reordered representation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute correct gradient	y = 2, residual -2: gradient [-8,-12], not [-12,-8].
2. Check coordinates	Step 0.1: a losses 3.24, 4.84 give -8; c losses 2.89, 5.29 give -12.
3. Apply correct update	At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Array **shape** and coordinate meaning both matter. A vector with two entries can have the right length but assign each derivative to the wrong parameter.

Label the entries a and c, then perturb one coordinate at a time to check their slopes. Repairing the coordinate mapping is necessary before using the gradient to update the network.

A correct gradient at one point does not establish every batch, **shape** or activation path is implemented correctly.

Change: The program changes its declared order to [c, a] and returns [-12,-8].

Prediction: That vector now matches the declared coordinate order.

Reason: Gradient values are meaningful only when their parameter names and array positions agree.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Shape	The sizes of an array along its declared axes
Contract	A stated condition that inputs and outputs must satisfy
Partial gradient	The change rate of a scalar loss with respect to one parameter while others are fixed
Regression check	A saved check rerun after a repair to detect the return of a defect

Repair from the first disagreement

TRACE STAGE FOR X [1 2]	EXPECTED RESULT	SUBMITTED RESULT
Input and dimensions	x length 2 W 3 by 2 b length 3	Dimensions match
Scores before ReLU	[1, 2, 0]	[1, 2, -1]
Hidden activations	[1, 2, 0]	[1, 2, 0]

The submitted third score omits bias 1. ReLU hides the error for this input, so matching final activations are not enough. At x = [3, 1], the same omission changes the third activation from 3 to 2.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Repair from the first disagreement

TRACE STAGE FOR X [1 2]	EXPECTED RESULT	SUBMITTED RESULT
Input and dimensions	x length 2 W 3 by 2 b length 3	Dimensions match
Scores before ReLU	[1, 2, 0]	[1, 2, -1]
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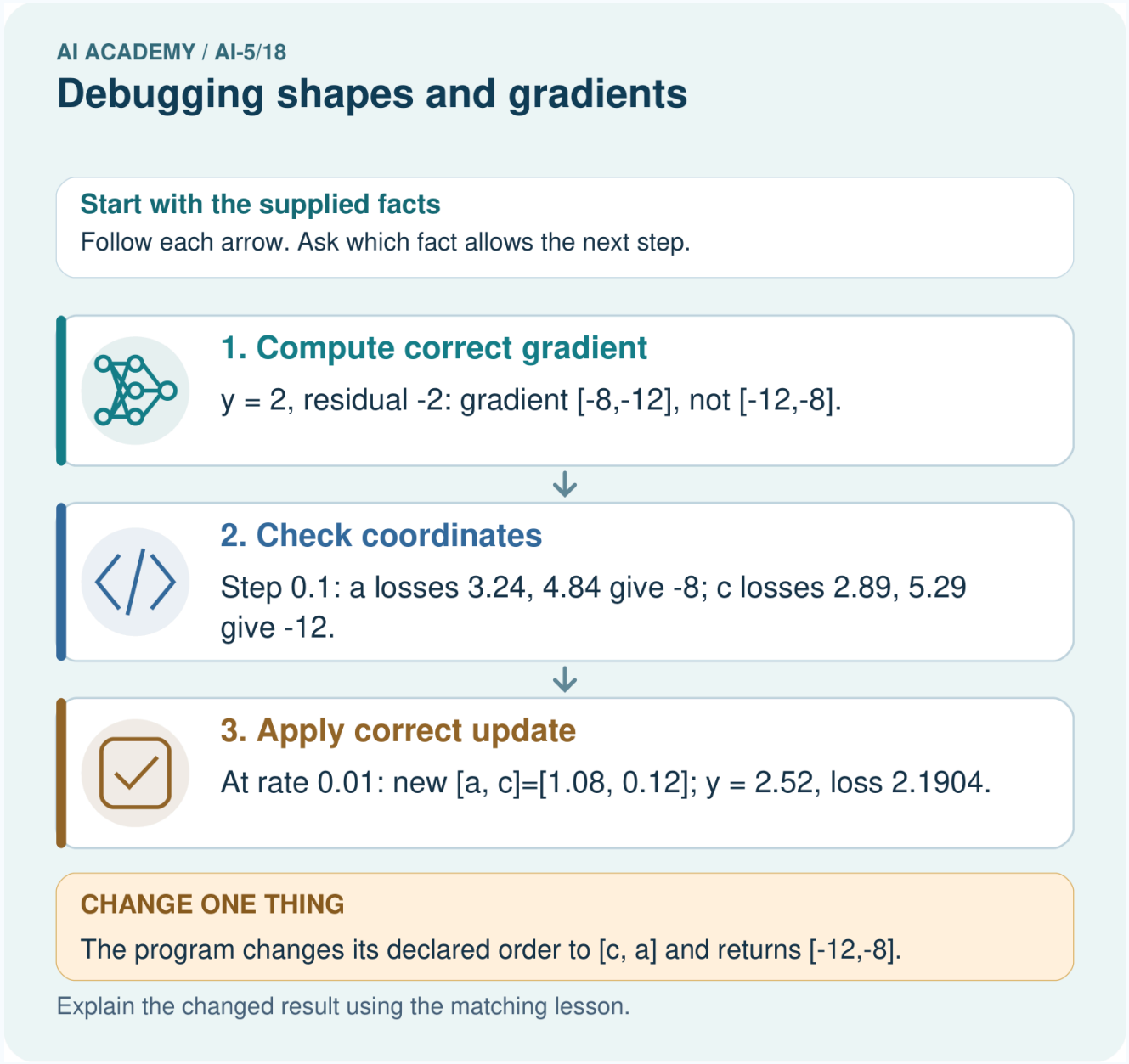
Trace x = [0, 2] with the correct layer. Then omit the third bias. Which scores differ, and why do the activation vectors still agree?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Debugging shapes and gradients

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Compute scores and activations for x = [3, 1] with the lesson layer, then omit the third bias. Identify the first discrepancy and the resulting activation error.

Teacher check: Teacher explanation: Correct scores and activations are [3, 1, 3]. Omitting the third bias produces [3, 1, 2]. The third score is the first discrepancy and now survives ReLU.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Debugging **shape**s and gradients. Proposed explanation: “Matching **output shape** guarantees the computation”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Correct dimensions can hide swapped features, missing biases or wrong axis meanings.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

What should a debugging record pair with every gradient value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Parameter name and index, expected **shape**, analytic derivative and numerical check; correct [a, c] gradient is [-8,-12]. Changed-case check: That vector now matches the declared coordinate order.

Gradient values are meaningful only when their parameter names and array positions agree.

Changed-case reasoning: That vector now matches the declared coordinate order. Gradient values are meaningful only when their parameter names and array positions agree.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide labelled columns for **input**s, scores, activations and parameter coordinates; allow calculators.

Extension: test several positive and negative **input**s against an independent explicit formula and explain why a test at a ReLU corner needs different reasoning from a smooth gradient check.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the difference between repeated and independently checked results; label all axes.
7 to 18 minutes	Trace both batch examples and use W1 to reject a dimension mismatch.
18 to 32 minutes	Run or trace the guarded dense-layer function and test invalid dimensions.
32 to 45 minutes	Use the visual and W2 to show how ReLU can hide a bias omission on one input.
Session 2 start to 8 minutes	Complete the guided x = [0, 2] trace, comparing scores with activations.
8 to 20 minutes	Probe w and b separately in the scalar diagnostic, then complete W3.
20 to 35 minutes	Collect W4 independently with four probe losses, ordered gradients and the corrected update.
35 to 45 minutes	Use W5 and the exit response to retain a regression checklist; administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main scores and exposed/hidden-bias checks supplied. W2[3, 1]correct[3, 1, 3], zeroedthirdbias[3, 1, 2]; firsterroratthirdscore. W3 diagnosticcorrectsubtract[1.2, .1], loss.25 versusaddloss2.25. W5 checkintermediates, malformeddimensions, andpositive-bias-sensitiveinput[3, 1]. Independenty3a+c, t2, old[0, 1]:aprobeloss 1.69/.49→−6, cprobe 1.21/.81→−2; swappedreportedentriesrepairto[−6, −2].

Rate.05 new[.3, 1.1], y2, loss 0. Keep coordinate order and saved old state explicit.

Diagnose the current misconceptions

Matching **output shape** guarantees the computation: Correct dimensions can hide swapped features, missing biases or wrong axis meanings.

Matching final activations proves all scores: ReLU can map different negative scores to the same zero.

Decreasing loss proves a correct gradient: An incorrect direction can still improve one case; compare coordinate probes and exact update expectations.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use y = 3a + c, target 2, loss (y - 2) squared, at [a, c] = [0, 1]. A program returns gradient [-2, -6] in order [a, c].

Use symmetric step 0.1 to check each coordinate, then update both with the corrected gradient and rate 0.05. Report new parameters and loss.

Teacher explanation: For a, losses at -0.1 and 0.1 are 1.69 and 0.49, giving -6. For c, losses at 0.9 and 1.1 are 1.21 and 0.81, giving -2. The entries were swapped. Correct update gives [0.3, 1.1], **prediction** 2 and loss 0.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Reject a shape mismatch

The layer expects W **shape** 3 by 2, b length 3 and x length 2. A submitted x has length 3. Should the last feature be silently ignored? State the failed **contract** and repair **process**.

Teacher explanation: No. Each weight row must have the same length as x.

Confirm the intended feature schema, then supply a valid two-feature **input** or deliberately redesign and retrain the layer; do not silently discard a value.

W2 Expose a hidden bias error

Compute scores and activations for x = [3, 1] with the lesson layer, then omit the third bias. Identify the first discrepancy and the resulting activation error.

Teacher explanation: Correct scores and activations are [3, 1, 3]. Omitting the third bias produces [3, 1, 2]. The third score is the first discrepancy and now survives ReLU.

W3 Check an update sign

For the diagnostic model at [1, 0], gradient [-4, -2] and rate 0.05, calculate the correct new parameters and loss. Compare with adding the gradient step.

Teacher explanation: Correct subtraction gives [1.2, 0.1], **prediction** 2.5 and loss 0.25. Addition gives [0.8, -0.1], **prediction** 1.5 and loss 2.25, compared with starting loss 1.

W5 Keep independent regression checks

A program passes only the x = [1, 2] final-activation test. Propose one intermediate-value check, one malformed-**input** check and one additional **input** that exposes the bias omission.

Teacher explanation: Check scores [1, 2, 0] before ReLU; reject a wrong **input** length or bias count; test x = [3, 1], whose third activation must be 3.

These checks address different defects rather than repeating the same final **output**.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: correct scores [0, 2, -1], activations [0, 2, 0]. With the third bias omitted, scores become [0, 2, -2], but activations stay [0, 2, 0]. Exit: neither check is sufficient.

Verify the intermediate score, coordinate meanings and expected update, use multiple independent **input**s and retain malformed-dimension checks.

W1 Reject a shape mismatch

W2 Expose a hidden bias error

W3 Check an update sign

W4 Debug a new gradient independently

Use y = 3a + c, target 2, loss (y - 2) squared, at [a, c] = [0, 1]. A program returns gradient [-2, -6] in order [a, c].

Use symmetric step 0.1 to check each coordinate, then update both with the corrected gradient and rate 0.05. Report new parameters and loss.

Teacher explanation: For a, losses at -0.1 and 0.1 are 1.69 and 0.49, giving -6. For c, losses at 0.9 and 1.1 are 1.21 and 0.81, giving -2. The entries were swapped. Correct update gives [0.3, 1.1], **prediction** 2 and loss 0.

W5 Keep independent regression checks

Independent assessment and fresh retry

W4 requires partials [-6, -2], new parameters [0.3, 1.1] and loss 0. Fresh retry: y = 2p + q, target 4, at [1, 1]. With step 0.1, p losses 1.44 and 0.64 give -4; q losses 1.21 and 0.81 give -2.

Rate 0.1 gives [1.4, 1.2], **prediction** 4 and loss 0. Require parameter ordering to remain explicit.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled debugging **shape**s and gradients **evidence** sheet to Image-pattern Network Investigation. Show how you can check dimension **contract**s before computing a layer.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide labelled columns for **input**s, scores, activations and parameter coordinates; allow calculators.

Extension: test several positive and negative **input**s against an independent explicit formula and explain why a test at a ReLU corner needs different reasoning from a smooth gradient check.

Week 19 explores learned representations and the distinctions hidden activations preserve.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/19, ai-5/21, ai-5/24. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Check dimension **contract**s before computing a layer

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Compute scores and activations for x = [3, 1] with the lesson layer, then omit the third bias. Identify the first discrepancy and the resulting activation error.

Reasoned practice answer: Teacher explanation: Correct scores and activations are [3, 1, 3]. Omitting the third bias produces [3, 1, 2]. The third score is the first discrepancy and now survives ReLU.

Check every learner before discussion: Use the worked case for Debugging **shape**s and gradients. Proposed explanation: “Matching **output shape** guarantees the computation”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Correct dimensions can hide swapped features, missing biases or wrong axis meanings.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The program changes its declared order to [c, a] and returns [-12,-8]. Require a **prediction** and a reason before comparing. Expected result: That vector now matches the declared coordinate order.

Reason: Gradient values are meaningful only when their parameter names and array positions agree.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

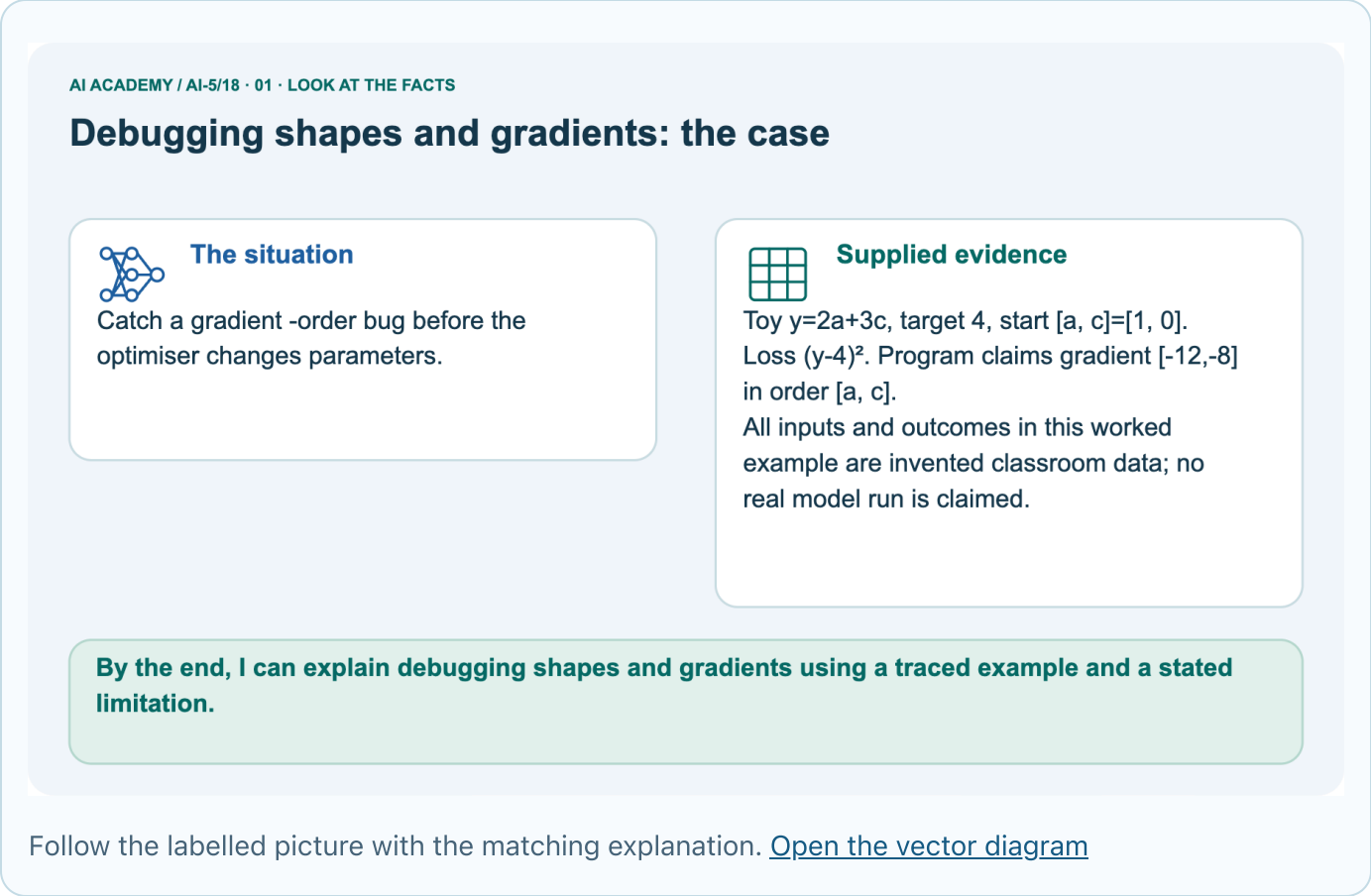
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Debugging shapes and gradients: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

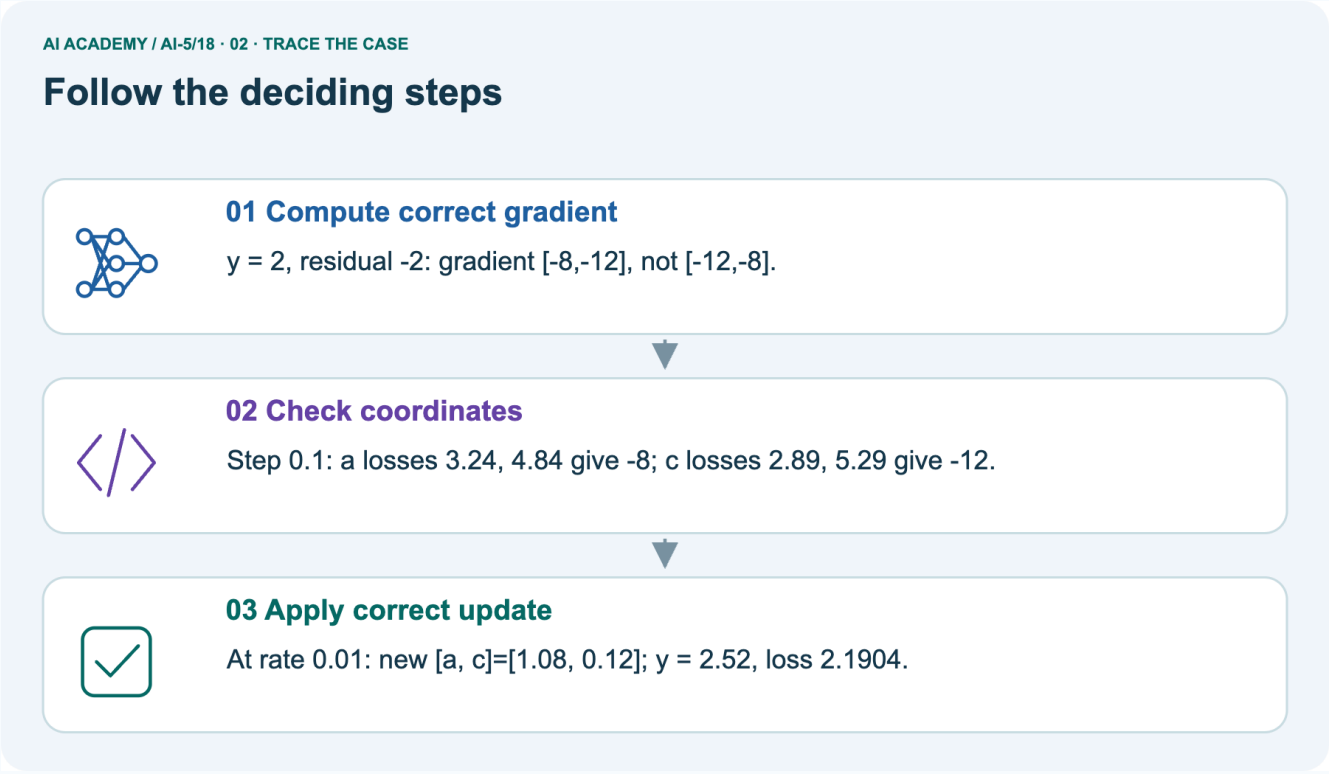
Catch a gradient -order bug before the optimiser changes parameters. Supplied facts: Toy y=2a+3c, target 4, start [a, c]=[1, 0]. Loss (y-4)². Program claims gradient [-12,-8] in order [a, c].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute correct gradient: y = 2, residual -2: gradient [-8,-12], not [-12,-8]. → Check coordinates: Step 0.1: a losses 3.24, 4.84 give -8; c losses 2.89, 5.29 give -12. → Apply correct update: At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Repair from the first disagreement

AI ACADEMY / AI-5/18 · 03 · INSPECT THE LESSON MODEL

Repair from the first disagreement

Trace stage for x [1 2]	Expected result	Submitted result
Input and dimensions	x length 2 W 3 by 2 b length 3	Dimensions match
Scores before ReLU	[1, 2, 0]	[1, 2, -1]
Hidden activations	[1, 2, 0]	[1, 2, 0]

The submitted third score omits bias 1. ReLU hides the error for this input, so matching final activations are not enough. At x = [3, 1], the same omission changes the third activation from 3 to 2.

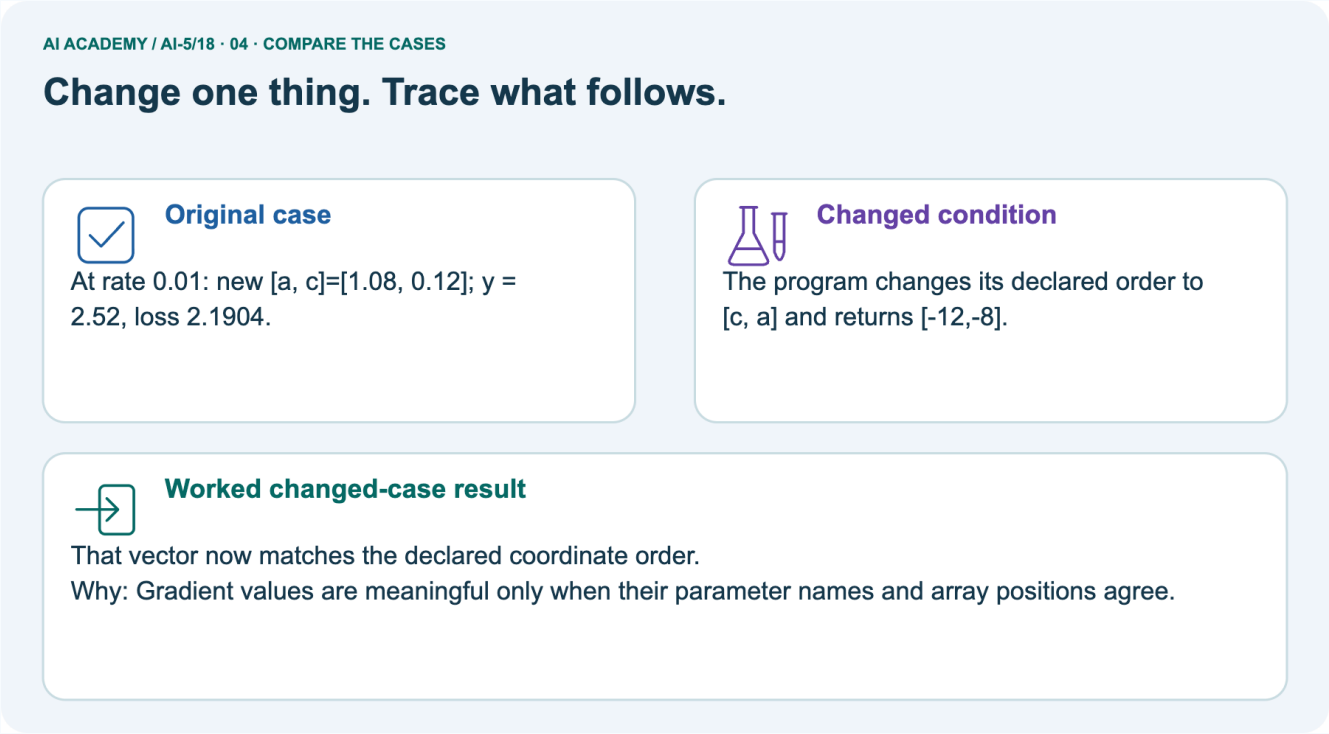
Follow the labelled picture with the matching explanation. Open the vector diagram

Repair from the first disagreement. **Input** and dimensions | x length 2 W 3 by 2 b length 3 | Dimensions match. Scores before ReLU | [1, 2, 0] | [1, 2, -1]. Hidden activations | [1, 2, 0] | [1, 2, 0]

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904. Changed condition: The program changes its declared order to [c, a] and returns [-12,-8].

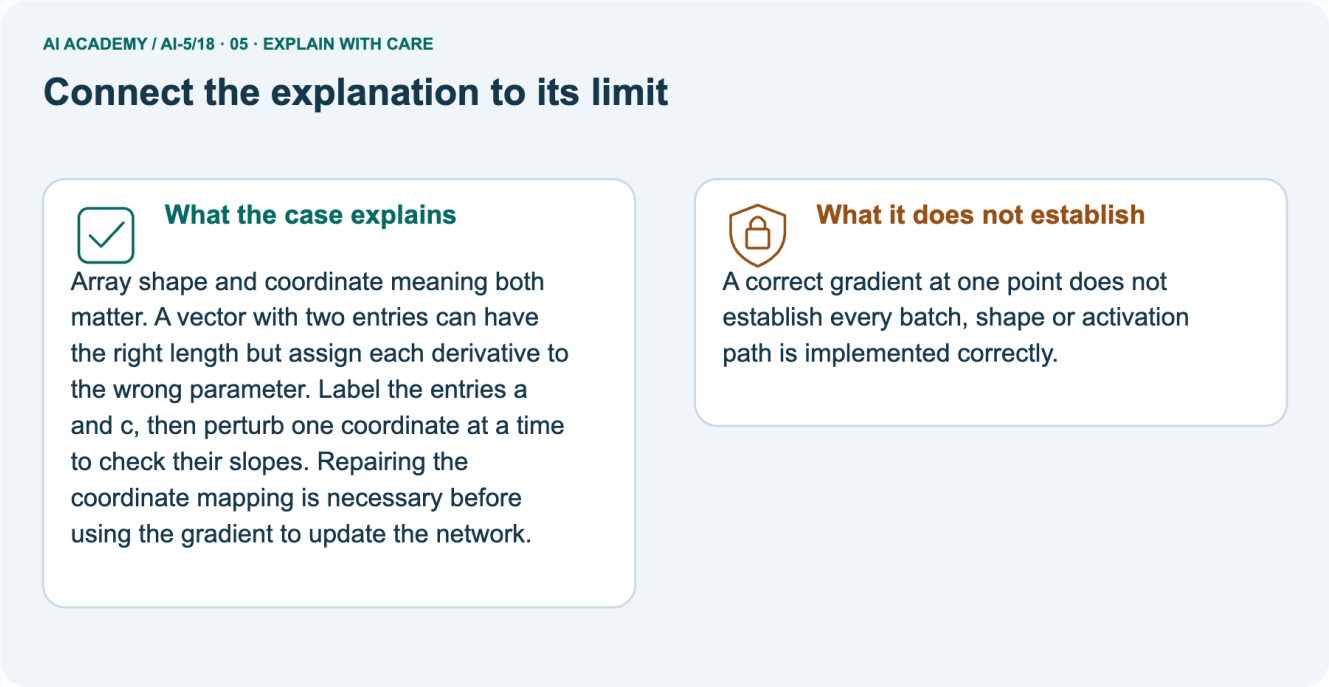
Expected result: That vector now matches the declared coordinate order. Reason: Gradient values are meaningful only when their parameter names and array positions agree.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Gradient values are meaningful only when their parameter names and array positions agree.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Array **shape** and coordinate meaning both matter. A vector with two entries can have the right length but assign each derivative to the wrong parameter.

Label the entries a and c, then perturb one coordinate at a time to check their slopes. Repairing the coordinate mapping is necessary before using the gradient to update the network.

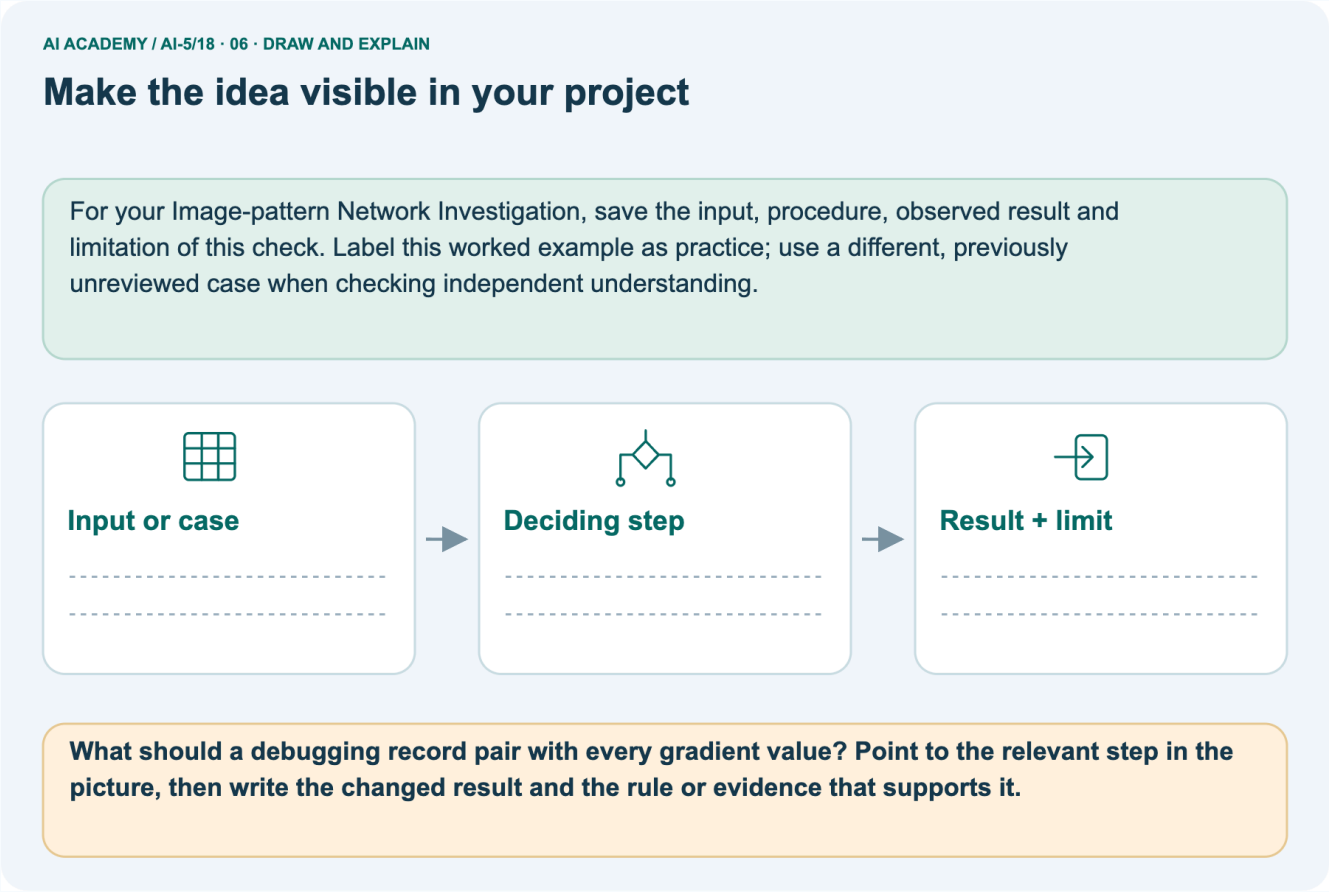
Limit: A correct gradient at one point does not establish every batch, **shape** or activation path is implemented correctly.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A correct gradient at one point does not establish every batch, **shape** or activation path is implemented correctly.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should a debugging record pair with every gradient value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Parameter name and index, expected **shape**, analytic derivative and numerical check; correct [a, c] gradient is [-8,-12]. Changed-case check: That vector now matches the declared coordinate order.

Gradient values are meaningful only when their parameter names and array positions agree.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Debugging shapes and gradients: the case

02. Follow the deciding steps

03. Repair from the first disagreement

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
</details>